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Test Name: Mock Test

Taken On: 16 Oct 2024 07:11:24 IST

Time Taken: 26 min 42 sec/ 28 min

Invited by: Ankush

Invited on: 16 Oct 2024 07:11:16 IST

Skills Score:

Tags Score:

- Algorithms 100/100
- Core CS 100/100
- Graph Theory 100/100
- Medium 100/100
- problem-solving 100/100

100%

100/100

scored in **Mock Test** in 26 min
42 sec on 16 Oct 2024 07:11:24
IST

Recruiter/Team Comments:

No Comments.

	Question Description	Time Taken	Score	Status
Q1	Breadth First Search: Shortest Reach > Coding	26 min 32 sec	100/ 100	✓

QUESTION 1

Correct Answer

Score 100

Breadth First Search: Shortest Reach > Coding

Graph Theory Algorithms Medium

problem-solving Core CS

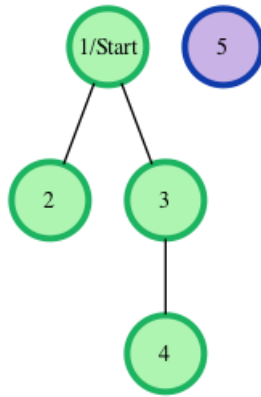
QUESTION DESCRIPTION

Consider an undirected graph where each edge weighs 6 units. Each of the nodes is labeled consecutively from 1 to n.

You will be given a number of queries. For each query, you will be given a list of edges describing an undirected graph. After you create a representation of the graph, you must determine and report the shortest distance to each of the other nodes from a given starting position using the *breadth-first search* algorithm (BFS). Return an array of distances from the start node in node number order. If a node is unreachable, return **−1** for that node.

Example

The following graph is based on the listed inputs:



$n = 5$ // number of nodes
 $m = 3$ // number of edges
 $edges = [1, 2], [1, 3], [3, 4]$
 $s = 1$ // starting node

All distances are from the start node **1**. Outputs are calculated for distances to nodes **2** through **5**: **[6, 6, 12, -1]**. Each edge is **6** units, and the unreachable node **5** has the required return distance of **-1**.

Function Description

Complete the `bfs` function in the editor below. If a node is unreachable, its distance is **-1**.

`bfs` has the following parameter(s):

- `int n`: the number of nodes
- `int m`: the number of edges
- `int edges[m][2]`: start and end nodes for edges
- `int s`: the node to start traversals from

Returns

`int[n-1]`: the distances to nodes in increasing node number order, not including the start node (-1 if a node is not reachable)

Input Format

The first line contains an integer **q**, the number of queries. Each of the following **q** sets of lines has the following format:

- The first line contains two space-separated integers **n** and **m**, the number of nodes and edges in the graph.
- Each line **i** of the **m** subsequent lines contains two space-separated integers, **u** and **v**, that describe an edge between nodes **u** and **v**.
- The last line contains a single integer, **s**, the node number to start from.

Constraints

- $1 \leq q \leq 10$
- $2 \leq n \leq 1000$
- $1 \leq m \leq \frac{n \cdot (n-1)}{2}$
- $1 \leq u, v, s \leq n$

Sample Input

```

2
4 2
1 2
1 3
1
3 1
2 3
2

```

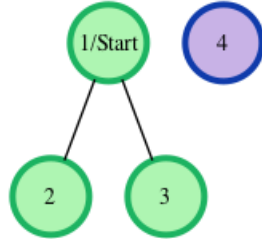
Sample Output

```
6 6 -1
-1 6
```

Explanation

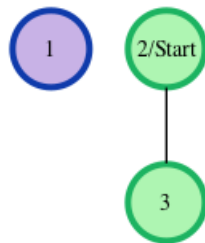
We perform the following two queries:

1. The given graph can be represented as:



where our *start* node, *s*, is node **1**. The shortest distances from *s* to the other nodes are one edge to node **2**, one edge to node **3**, and an infinite distance to node **4** (which it is not connected to). We then return an array of distances from node **1** to nodes **2**, **3**, and **4** (respectively): **[6, 6, -1]**.

2. The given graph can be represented as:



where our *start* node, *s*, is node **2**. There is only one edge here, so node **1** is unreachable from node **2** and node **3** has one edge connecting it to node **2**. We then return an array of distances from node **2** to nodes **1**, and **3** (respectively): **[-1, 6]**.

Note: Recall that the actual length of each edge is **6**, and we return **-1** as the distance to any node that is unreachable from *s*.

CANDIDATE ANSWER

Language used: **Python 3**

```
1
2 #
3 # Complete the 'bfs' function below.
4 #
5 # The function is expected to return an INTEGER_ARRAY.
6 # The function accepts following parameters:
7 # 1. INTEGER n
8 # 2. INTEGER m
9 # 3. 2D_INTEGER_ARRAY edges
10 # 4. INTEGER s
11 #
12
13
14 def shift_right(heap, idx):
15     hlen = len(heap)
16
17     while True:
18         curr, left, right = idx, idx * 2 + 1, idx * 2 + 2
19
20         if left < hlen and heap[curr] > heap[left]:
21             curr = left
22         if right < hlen and heap[curr] > heap[right]:
```

```

23         curr = right
24
25         if curr != idx:
26             heap[curr], heap[idx] = heap[idx], heap[curr]
27             idx = curr
28         else:
29             break
30
31 def shift_left(heap, idx):
32     item = heap[idx]
33
34     while idx > 0:
35         left = (idx - 1) >> 1
36         if heap[left] > item:
37             heap[idx] = heap[left]
38             idx = left
39         else:
40             heap[idx] = item
41             break
42
43 def pop(heap):
44     hn = len(heap)
45
46     if hn == 1:
47         return heap.pop()
48
49     item = heap[0]
50     heap[0] = heap.pop()
51     shift_right(heap, 0)
52
53     return item
54
55 def push(heap, item):
56     heap.append(item)
57     shift_left(heap, len(heap)-1)
58
59 def find_min(n, distance, visited):
60     m = sys.maxsize
61     k = -1
62
63     for i in range(n):
64         if visited[i] is False and m > distance[i]:
65             m = distance[i]
66             k = i
67
68     return k
69
70 def bfs(n, m, edges, s):
71     # Write your code here
72     adj = [] * n
73
74     for _ in range(n):
75         adj.append([] * n)
76
77     for e in edges:
78         adj[e[0] - 1].append(e[1] - 1)
79         adj[e[1] - 1].append(e[0] - 1)
80
81     max_edges = len(edges) + 1
82     visited = [False] * n
83     distance = [max_edges] * n
84
85     distance[s - 1] = 0

```

```

86
87     for _ in range(n):
88         b = find_min(n, distance, visited)
89
90
91         if b == -1:
92             break
93
94         visited[b] = True
95
96         for t in adj[b]:
97             if visited[t] is False and distance[t] > distance[b] + 1:
98                 distance[t] = distance[b] + 1
99
100
101     r = []
102     for i in range(n):
103         if i != s-1:
104             r.append(distance[i] * 6 if distance[i] < max_edges else -1)
105
106     return r
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TESTCASE	DIFFICULTY	TYPE	STATUS	SCORE	TIME TAKEN	MEMORY USED
Testcase 1	Easy	Sample case	✔ Success	0	0.0903 sec	14.7 KB
Testcase 2	Medium	Hidden case	✔ Success	5	0.0938 sec	14.8 KB
Testcase 3	Medium	Hidden case	✔ Success	5	0.1826 sec	16.9 KB
Testcase 4	Hard	Hidden case	✔ Success	15	0.0822 sec	14.8 KB
Testcase 5	Hard	Hidden case	✔ Success	15	0.0968 sec	14.7 KB
Testcase 6	Hard	Hidden case	✔ Success	30	0.7777 sec	32.8 KB
Testcase 7	Hard	Hidden case	✔ Success	30	0.2131 sec	15.1 KB
Testcase 8	Easy	Sample case	✔ Success	0	0.0873 sec	14.8 KB

No Comments